सत्यं शिवम्सुन्दरम्

WEST BENGAL STATE UNIVERSITY

B.Sc. Major 1st Semester Examination, 2024-25

CMSDSC101T-COMPUTER SCIENCE (MAJOR)

Time Allotted: 2 Hours

Full Marks: 50

The figures in the margin indicate full marks.
Candidates should answer in their own words and adhere to the word limit as practicable.
All symbols are of usual significance.

1. Answer any *five* questions from the following: 2×5 = 10

(a) What is the difference between for and while statement?

(b) What do you mean by nested loops?

(c) What is a static variable? Give an example.

(d) What are arrays? How arrays are declared?

(e) What is function in C?

(f) What is string? How to declare a string in C?

(g) What are the different format specifiers in C?

(h) What will be the output of the following code?

```
Void main()
{
printf{"nn \n\n nn\n"};
printf{"nn/n/n nn/n"};
}
```

Answer any *five* questions from the following 8×5 = 40

2. What is the difference between continue and break statements in C? Give example. Write a program to print sum of the following series: 4+4

$$1+\frac{1}{2}+\frac{1}{3}+\cdots+\frac{1}{n}$$

The value of n should be given through terminal.

3. Write the difference between call by value and call by reference. Give programming example. Write the function of size of operator in C. (3+3)+2

4. What is a 'struct' in C, and how does it differ from an array? What is 'const' keyword in C? Briefly describe all types of storage class in C. 2+2+4

5. (a) What is a NULL Pointer? Whether it is same as an uninitialized pointer? 3+3+2
   (b) Difference between formal argument and actual argument.
   (c) What is pointer to a pointer?

6. (a) How can we check whether the contents of two structure variables are same or not? Explain with a program code. 4+2+2
   (b) What is the difference between null array and an empty array?
   (c) What is the use of typedef?

7. (a) Write a C program to find out the sum of diagonal elements of a matrix. 6+2
   (b) What do you mean by memory leak?

8. (a) How to switch the values of two variable without using third variable? Show with the help of a program code. 3+5
   (b) Write a C program to compare two strings using pointers.

9. (a) What are header files? Can I run program without using header file? 3+2+3
   (b) What is the difference between getch() and getchar()?
   (c) What are the stdin and stdout file? What are binary files?

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